



An approach to finite element modeling of liquid storage tanks in ANSYS: A review

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Abstract

Post-earthquake utilities of liquid storage tanks do not need any over emphasis. Liquid storage tanks used in public water distribution system provide water for firefighting and also for drinking and household purposes. Tanks used in industry are equally critical as they contain hazardous liquids and their leakage may lead to catastrophic situation. Thus, seismic safety of liquid storage tanks, which are life line structures, is very important. The prime aim of the present work is to review the sloshing phenomenon studied by various researchers in the past. The study is broadly divided into two parts namely tanks without obstruction and tanks with obstruction and its effects on sloshing parameters. A brief review on sloshing problem in complex tank geometries in form of Intze, elliptical and conical tanks is also presented. The study also presents an overview and development of mathematical formulations to predict the sloshing parameters in form of sloshing frequency, convective mass excited and hydrodynamic pressure profiles which enable present structural engineers in designing earthquake-resistant storage structures. A brief review on modeling circular and rectangular liquid storage tanks using ANSYS software is also presented along with the assumptions and suitable element for modeling. The study also investigates the advantages and disadvantages of the different analysis methods available in the finite element program (ANSYS) to obtain the sloshing parameters. The study concludes that an optimum mesh size as detailed in the present study must be evaluated to determine slosh parameters. An arbitrary fine mesh results in a limited number of modes extracted from the finite element analysis and further gives inaccurate solutions.

Keywords Liquid storage tanks · Sloshing frequency · Internal obstruction · Hydrodynamic pressure

Introduction

Seismic design of liquid storage tanks requires knowledge of sloshing frequency of liquid and hydrodynamic pressure on the tank wall. This further requires proper analysis of fluid–structure interaction under seismic excitation. During the early 1960's, the work done by Housner [1] has proved to be pivotal in analyzing liquid storage tanks. In the field of earthquake engineering, the sloshing problem has been simplified with the help of mechanical analogue developed by Housner. The model as shown in Fig. 1 divides liquid into impulsive and convective parts. The study is helpful in understanding dynamic response of ground-supported rectangular as well as cylindrical tanks. The liquid contained

in tanks is considered to be in viscid and incompressible. Figure 1 shows the idealized spring mass model for ground-supported tanks. During an earthquake, all the contents of tank or reservoir on the ground will be excited (sloshing phenomenon), moreover two forces are exerted on tank by water. First, when the walls of the tank accelerate back and forth a certain fraction of the water is forced to participate in this motion, which exerts a reactive force on the tank which is same as exerted by a mass m_i that is attached rigidly to the tank at the proper height as shown in Fig. 1 and is known as impulsive mass of liquid. Second is the motion of the tank walls excites the water into oscillations which in turn exert an oscillating force on the tank. This oscillating force is the same as would be exerted by a mass m_c that can oscillate horizontally against a restraining spring (k_c , which is spring stiffness of convective mode) and is known as convective mass of liquid. The height of convective and impulsive mass from bottom of tank is denoted by h_c and h_i , respectively.

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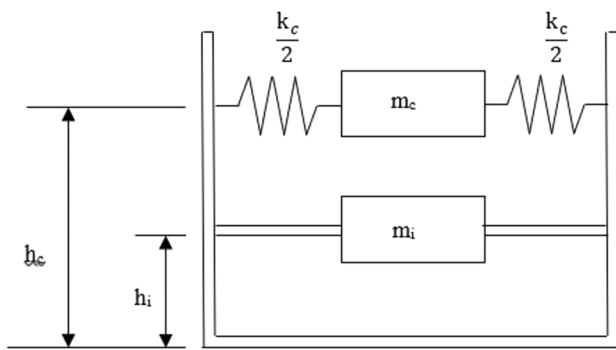


Fig. 1 Two-mass model for tank Housner (1963)



Fig. 2 Buckled water tank lifted off the base during Lander Earthquake (1992, California)

It has been observed that because of easy design, construction and ability to resist primary hydrostatic pressures, ground-supported cylindrical tanks are more numerous than any other configuration of liquid storage tanks. Many researchers have also studied the behavior and damage caused to these structures by strong ground motions. One of the most characteristic type of damage is the circumferential protrusion near the base, commonly referred to as “elephant’s foot” as shown in Fig. 2 that can form due to large axial compressive stresses caused by beam-like bending of the tank wall as studied by Koller and Malhotra [2]. Hydrodynamic forces developed within the tank may damage the inner columns supporting the roof. Excessive sloshing of the contained liquid may lead to buckling of the roof or upper portion of the tank wall infixed roof tanks and failure of the roof seals in floating roof tanks. Uplifting of the tank due to overturning moments generated by the hydrodynamic forces may damage the anchor bolts, mantle-base plate interface and piping connections. Overturning moments may also lead to differential settlement of the tank foundation as investigated by Barros [3]. The other significant damages incurred to these ground-supported storage tanks are in form of poor performance of floating roof systems and

fire related damages. The damages incurred to floating roof of Naphtha tanks due to fires during Kocaeli earthquake as cited by [4–6] have been investigated by Yazici and Cili [7]. The study reports the detrimental effects of sloshing phenomena on tank walls and recommends that the site-specific time history records to be employed in the critical design of such tanks.

For proper understanding and prediction of these failures, the fluid–structure interaction and sloshing behavior of liquid shall be critically analyzed. Nowadays, use of finite element to model liquid is becoming more and more popular (ANSYS software). The present study amasses the available literature highlighting the sloshing problem in liquid storage tanks along with the mathematical and analytical approach to calculate the parameters involved. The study also addresses the procedure to be followed in modeling and analyzing different types of liquid storage tanks in finite element program (ANSYS). The study also discusses the best suited method of free vibration analysis known as modal analysis to obtain the sloshing parameter results available in ANSYS software. The study also evaluates the importance of meshing in analyzing tank problem and concludes the merits and demerits of fine meshing of fluid in finite element package. The solution to obtain accurate results from free vibration method is also discussed. The importance of meshing based on the aspect ratio of tank geometry is also critical in obtaining accurate results. The issues encountered during fine meshing the fluid in ANSYS are also summarized. In the next section, a detailed review of ground-supported liquid storage tanks highlighting the sloshing problem is abridged.

Literature review

The analytical approach developed by various researchers to investigate the sloshing problem has enabled the structural engineers in accurately determining the hydrodynamic forces for circular and rectangular tanks. The dynamic response of regular liquid storage tank geometries has been studied extensively in past several decades. For tanks of other shapes, the sloshing frequency and hydrodynamic pressure equations are not available directly. Moreover, the configuration of the tanks for most of these studies comprised of the regular geometry like rectangular and circular. The study reviews the past methods elaborating the significance of seismic analysis and fluid–structure interaction of these special structures. The study is restricted to vertical-type tanks and studies on horizontally laid tanks are neglected. The review primarily discusses the sloshing phenomenon in the ground-supported tanks of regular geometry in depth with a brief introduction and challenges faced in the analysis and

design of complex tank geometries also investigated. The review is divided into following three categories:

- (i) Tanks without obstruction.
- (ii) Tanks with internal obstruction.
- (iii) Complex tank geometries.

Tanks without obstruction

The chronological development and advancement to analyze the regular shaped liquid storage tanks over the past few decades are summarized in this section. The study discusses the analytical approach and formulations to accurately predict the sloshing forces which the storage tanks are subjected in the advent of a natural calamity like an earthquake. The assumptions, validity and accuracy of the developed methods in comparison with the recently developed software-based approaches in form of computer-based finite element (FE) packages namely ANSYS, MATLAB, ADINA and ABAQUS are also presented. In the year 1949, Jacobsen [8] calculated the effective hydrodynamic masses and mass moments for the liquid inside a cylindrical tank and outside cylindrical pier when the base of the tank or pier experiences a horizontal acceleration. However, the study was restrained to impulsive hydrodynamic and convective effect is ignored. The analytical formulations for hydrodynamic pressure and force moments were derived along with effective masses and their locations. Given analysis was implemented in solving specific problem. Impulsive velocities and contour plots of their resultants were also studied. The landmark contributions by Housner [9] in the analysis of liquid storage tanks have enabled the present structural designers in accurately predicting the dynamic behavior of fluids contained in tanks. The study is separated in two parts for calculation of impulsive and convective parameters of fluids in tanks. The fluid is assumed incompressible and the fluid displacements are assumed to be small for derivation of generalized expressions for impulsive and convective pressures from basic principles. The expressions for sloshing frequencies, impulsive and convective masses and other hydrodynamic characteristics are also given. The study is further extended to formulate the parameters of mechanical models for circular and rectangular tanks and has helped in the formulation of various international codes of practice [10–12]. Arya et al. [13] studied the dynamic characteristics of fluid containers under various combinations of axial and circumferential waves. The study was carried out for cylindrical shells which were clamped at base and free at top. Both the conditions, i.e., tanks empty as well as filled to varying depths were analyzed using energy method. The effect of water has been considered by virtual mass factor. Veletsos and Yang [14] studied the hydrodynamic effects induced in rigid tanks including the comparison of exact solution with Housner's

approximate solution. A simple procedure was developed for assessing the effect of tank flexibility. The study also included the formulation and application of a more refined analysis in which tank was analyzed with the help of shell theory. Both empty and liquid-filled tanks were analyzed for obtaining natural frequencies and mode of vibration. Aslam et al. [15] studied the problem of sloshing both analytically and experimentally in annular tanks subjected to both harmonic and arbitrary ground motion. The main aim of the study was to determine the time history of water surface displacements for an applied ground motion in boiled water reactor. Experimentally, the response characteristics were studied on small-scale model subjected to ground motion with the help of shake table. Haroun et al. [16] developed a simple and computationally effective method for computing the dynamic characteristics of ground-supported cylindrical liquid storage tanks. The significance of soil structure interaction for response of anchored tanks was also studied in brief. Moreover, the convective pressure can be evaluated by considering tank wall to be rigid whereas the impulsive pressure can be determined by analyzing the liquid shell system and neglecting sloshing motion. The validity of the method of analysis has been confirmed by both scale method testing as well as full-scale field testing. In 1984, Veletsos [17] developed design guide for engineers from the ground motion studies of liquid storage tanks. The document provides mathematical formulations for calculation of pressure forces considering the flexibility and rigidity of tank in the first three modes of vibrations and refined the expression given by Housner. The study also illustrates model for safe construction and safety of steel and concrete liquified natural gas tanks. Stuart et al. [18] demonstrated the effectiveness and reliability of finite element method-based techniques to evaluate the fluid–structure interaction effects on high-level waste storage tanks (HLWST). A systematic step by step approach for dynamic analysis of HLWST using ADRINA fluid elements has been presented. The results obtained for modal analysis were compared with available data whereas that obtained from dynamic response of tank were verified for displacements and hydrodynamic pressure. Malhotra [19] studied the seismic response of unanchored liquid storage tanks, supported directly on flexible soil foundations. An efficient method was presented for the dynamic response analysis of flexibly supported unanchored tanks. The study showed that the flexibility of the foundation reduces the overturning base moment and significantly reduces the axial compressive stresses in tank wall. It was also concluded the hydrodynamic base pressures reduce the uplifting resistance and damping of the base plate. Further, Malhotra et al. [20] proposed a simple procedure adopted by Eurocode 8 considering the impulsive and sloshing behavior in elastic steel and concrete liquid storage tanks rigidly fixed at base. The study also includes a detailed analytical problem calculating the

modal and seismic responses of a steel tank rigidly fixed to a concrete foundation. Experimental tests by Gardarsson and Yeh [21] on rectangular tanks with vertical walls and sloping bottoms showed the phenomenon of hardening spring and softening spring, respectively. The study also showed the phenomena of wave breaking at resonance and hence the available analytical solutions were not applicable to validate the results. Jaiswal et al. [22] conducted experimental and numerical study to validate the analytical results for first sloshing (convective) frequency. The experimental results for tank shapes such as conical and truncated pyramid type of geometry were also conducted. The effect of a centrally placed obstruction in square tanks on the first sloshing frequency was found to decrease the sloshing frequency as the size of obstruction increases. Tang et al. [23] studied the dynamic response of liquid in upright annular rigid cylindrical tanks subjected to the horizontal seismic excitation at its base. Analytical method was employed to the study the response. A simple procedure for computing the seismic response of annular cylindrical tanks considering the effect of tank wall flexibility was also proposed. The response quantities evaluated in the study included hydrodynamic pressure, base shear, base moment, sloshing frequencies and surface displacement. Graphical presentation of these quantities in terms of impulsive and convective components has been proposed. Ghaemmghami and Kianoush [24] used finite element method to investigate seismic behavior of rectangular tanks in 2-D space. The method considered both impulsive and convective responses of the tank system. The results show that wall flexibility and fluid damping properties have a major effect on seismic behavior of tanks. The results were verified by numerical methods and other available methods available. Moslemi and Kianoush [25] conducted dynamic analysis in the form of time history on ground-supported circular tanks using ANSYS software. The effect of tank wall flexibility, vertical component of earthquake, sloshing of liquid free surface, tank aspect ratio and base fixity on dynamic behavior was investigated. The study concluded that the wall flexibility should be taken into consideration in seismic design of the tanks due a significant increase in the impulsive part of response. Recently, Oliaiy et al. [26] evaluated the efficiency of padded walls in reducing the sloshing effects in rectangular tanks using analytical approach. Jung et al. [27] investigated the sloshing phenomenon in rectangular tanks using finite volume method.

The other notable research directed toward the earthquake response of liquid storage tanks experimentally and analytically using finite element approach has been performed by Tedesco [28], Kianoush and Ghaemmghami [29], Choun [30], Kotrasova and Grajciar [31], Vathi and Karamanos [32], Malhotra et al. [33], Sivy et al. [34] and Otremba and Navarrete [35]. Recently, Zhao et al. [36] validated the numerical sloshing pressure histories with

the available experimental studies on liquified natural gas rectangular tanks subjected to forced vibrations. The study also investigated the effects of viscosity on the sloshing problem based on the artificial damping models as proposed by Ning et al. [37, 38]. The rectangular tanks subjected to linear and nonlinear sloshing phenomenon have been investigated by [39–41]. However, the extensive work by Moslemi et al. [42] advocates the codes of practice to include the corner sloshing problem especially in shallow tanks as compared with tall tanks. The study also proposed that the nonlinear sloshing problem leads to insufficient free board and is a source of roof damage particularly in rectangular concrete structures. The comparison of various sloshing parameters has been investigated using ABAQUS/Explicit package. Research directed toward validating the sloshing parameters of ground-supported rectangular and cylindrical liquid storage tanks using finite element packages with the available analytical formulations has been investigated by [43–48]. An extensive FE investigation is also performed by Liu [49] focusing on the buckling problems in liquid storage tanks resulting from the seismic excitations and proposed that the use of fluid–structure interaction method taking in account the material and geometrical nonlinearity of the structure. The other significant contributions in evaluating the behavior of the liquid storage with the help of fluid–structure interaction method have been studied by [50–60]. Thus, it is observed from the above-mentioned recent studies that the reliability on computer-aided software packages has increased due to the easy in solving complex sloshing problems within stipulated time. However, it is essential to validate the models with the available expressions for sloshing parameters for different tank geometries. The present study summarizes the available sloshing frequency and sloshing mass ratio equations as proposed by Housner and NZS 3106:1986 for rectangular tanks are represented by Eqs. 1 and 2, respectively. The nomenclature shows that “ m_c ” is convective mass excited in fundamental mode, “ m ” is total mass of liquid, “ H ” is height and “ L ” is the length of the tank along the direction in which mode is considered. In the expressions 1 and 2, f_i is the sloshing frequency of i^{th} mode in Hz, l is the half the length of wall, “ H ” is the height of liquid and the constant α_i is equal to $\pi/2$, $3\pi/2$, $5\pi/2$ for the first three modes. The present study showcases the need to develop mathematical equations to determine sloshing parameters in rectangular tanks with obstructions at higher modes as developed for cylindrical tanks and discussed in next section. The expressions for sloshing frequency and sloshing mass excited in cylindrical tanks as proposed by various researchers are tabulated in Tables 1 and 2. The convective mass for fundamental mode can be found out by analytical formula given by Housner (1963).

Table 1 Expressions for convective frequency for circular tanks without obstruction

Study	Convective sloshing frequency
Housner [1] (1963)	$f = \frac{1}{2\pi} \sqrt{\sqrt{\frac{27}{8}} \frac{g}{r} \tanh \sqrt{\frac{27}{8}} \frac{h}{r}}$ In the above equation, f is the first mode sloshing frequency in Hz g is the acceleration due to gravity in m/sec² h = Height of tank (m) r = Radius of tank in m
NASA SP-8009 [61] (1968)	$\omega_n = \sqrt{\epsilon_n g a^{-1} \tanh(\epsilon_n h a^{-1})}$ where ω_n = Sloshing frequency of n^{th} mode, rad/sec g = Longitudinal acceleration, ft/sec² h = Height of quiescent fluid surface tank, ft a = Tank radius, ft ϵ_n = determined from root of first derivative of $J_1(x)$ known as the Bessel function of first kind and order having values equal to 1.841, 5.33 and 8.53 in first three modes, respectively
Veletsos and Yang [14] (1976)	$f_k = \frac{1}{2\pi} \sqrt{\lambda_k \frac{g}{a} \tanh(\lambda_k \frac{H}{a})}$ where f_k = k^{th} sloshing frequency in Hz g = Acceleration due to gravity in m/sec² h = Height of fluid in tank (m) a = Radius of cylindrical tank (m) λ_k = First derivative of the Bessel function of first kind and first order having values equal to 1.841, 5.33 and 8.53 in first three modes, respectively

Table 2 Expression for convective mass ratio for circular tanks without obstruction

Study	Convective mass ratio
Housner [1] (1963)	$\frac{m_c}{m} = 0.23 \frac{\tanh 3.68 \frac{h}{D}}{\frac{h}{D}}$ $\frac{m_c}{m}$ = convective mass ratio h = Height of tank (m) D = Diameter of tank (m)
Veletsos and Yang [14] (1976)	$\frac{m_k}{m} = \left[\frac{2}{\lambda_k^2 - 1} \frac{a}{\lambda_k H} \tanh \left(\lambda_k \frac{H}{a} \right) \right]$ m_k = mass of the k^{th} convective mode m = total mass of liquid in tank a = Radius of cylindrical tank H = Height of incompressible and inviscid liquid λ_k = First derivative of the Bessel function of first kind and first order having values equal to 1.841, 5.33 and 8.53 in first three modes, respectively

$$f_i = \frac{\sqrt{\alpha_i \tanh(\alpha_i H/l)}}{2\pi\sqrt{l/g}} \quad (1)$$

$$\frac{m_c}{m} = 0.264 \frac{\tanh\left(3.16 \frac{H}{L}\right)}{\frac{H}{L}} \quad (2)$$

In the last few years, the study on mitigating the sloshing phenomenon in the liquid storage tanks has been extensively studied by providing baffles in form of obstructions. The research directed toward the effects of obstruction on the sloshing phenomenon is discussed in the next section. The studies presented discuss the effectiveness of finite element method to analyze regular tanks geometries with the presence of internal obstructions along with tanks of different shapes and geometries. In many cases, vertical objects are present inside the container. For example, in water tanks, vertical columns supporting the roof are present inside the tank. In many tanks baffle walls are quite common. In nuclear reactors also, internals such as core support structures, grid plate, thermal baffles and inner vessels are present. Due to the presence of these objects, which will act as obstruction in the sloshing of liquid, the sloshing behavior will change which in turn will affect the hydrodynamic pressure/force on the tank wall. From seismic safety point of view, effect of these obstructions on sloshing characteristics shall be properly understood and is detailed herein.

Tanks with internal obstruction

The analytical studies evaluating the sloshing frequency and other sloshing parameters of cylindrical tank with internal obstruction was first developed by Abramson [62]. The study outlines the procedure to obtain the characteristics of lateral sloshing behavior, primarily liquid frequencies, mode shapes and forced response characteristics for various container configurations. The study considered only rigid tanks. For cylindrical tanks, eigenvalues of fluid are obtained using Eq. 4 from the free surface condition is given by Eq. 3

$$\frac{\partial^2 \varphi_1}{\partial t^2} + g \frac{\partial \varphi_1}{\partial z} = 0 \text{ at free surface } z = 0 \quad (3)$$

$$\omega_{nm}^2 = \frac{g}{a} \xi_{nm} \tanh\left(\xi_{nm} \frac{h}{a}\right) \quad m, n = 0, 1, 2 \quad (4)$$

where ω = natural circular frequency. g = gravitational acceleration. h = liquid depth. a = outer radius of tank. m, n = Integer ranging from 0 to ∞ . It should be noted that the values of constant ξ_{nm} are the positive roots of Eq. 5 for the case $m/2\alpha = 1.0$ and are plotted and evaluated from Fig. 3.

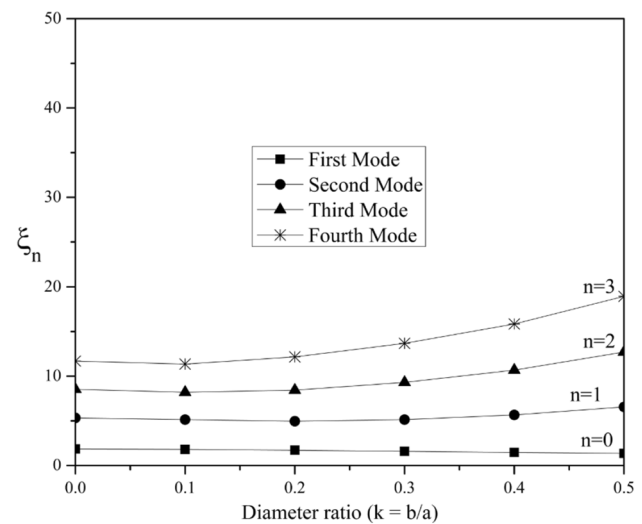


Fig. 3 Roots of $\Delta_1(\xi)=0$ versus parameter k (Abramson 1968)

$$\Delta_{m/2\alpha} = J'_{m/2\alpha}(\xi)Y'_{m/2\alpha}(k\xi) - J'_{m/2\alpha}(k\xi)Y'_{m/2\alpha}(\xi) = 0 \quad (5)$$

where k = diameter ratio of inner and outer wall tanks. b = inner radius of tank. α = Vertex angle of sector tank. $J'_{m/2\alpha}()$ = First derivative of Bessel function of first kind and order. $Y'_{m/2\alpha}()$ = First derivative of Bessel function of second kind and order.

Watson and Evans [63] solved a number of two-dimensional problems for resonant frequencies of oscillation of fluid in rectangular or circular containers having internal bodies. The internal bodies include surface or bottom mounted vertical blocks or circular apertures. An approximate formula for an arbitrary shaped body in container is used to solve the resonant frequencies of rectangular tank having an internal body as submerged cylinder. The results by the proposed method are found to be in good agreement with the more accurate Galerkin approach. It was also observed that in all cases as the size of internal body increased, resonant frequency decreased.

Jeyakumaran and Mciver [64] proposed an approximate method for calculation of oscillation frequencies of rectangular tank containing submerged cylinder of arbitrary cross section in equivalent three-dimensional motion. Effect of both vertical and horizontal thin baffles on frequencies was also studied. It was observed that the introduction of rigid internal structure reduces the frequency of fluid in rectangular tank. Drake [65] investigated theoretically the change in fundamental frequency due to the presence of an array of pipes in circular tanks. The problem was simplified to a two-dimensional problem. Special mathematical functions were used to describe boundary condition and also satisfy velocity potential of flow. A boundary element was used to investigate the case of a tank with an arbitrary

array of pipes and is verified by analytical solution for circular and annular tanks. The expression for natural frequency proposed by the study is given by Eq. 6. In terms, ω_n is the sloshing frequency of n th mode, rad/sec², g is the longitudinal acceleration, ft/sec², h is the height of tank, ft, a is the outer radius of tank, ft and b is the inner radius of tank, ft.

$$\omega_n = \sqrt{gk_n \tanh(k_n h)} \quad (6)$$

Moreover, the values of constant k_1 for different ratios of inner to outer radius in graphical form are as shown in Fig. 4.

It should be noted that the formula proposed by Drake is valid for first mode only and nothing about the higher modes is mentioned for annular tanks. Askari and Daneshmand [66] analytically investigated the effect of rigid internal body on coupled vibration of a partially filled cylindrical container. The rigid body was concentric and partially immersed inside the tank. The natural frequency and modes of cylindrical tank were calculated using Rayleigh–Ritz method. Galerkin method was used to solve velocity potential equations for liquid motion. Finite element method was used to validate the analytical method along with experimental studies. The effect of fluid level, radius of internal body and length of internal body on natural frequency was also investigated. Finally, it was concluded that the increase in radius of rigid body decreases sloshing frequency. The formula proposed for calculation of frequency of container partially filled with fluid is given by Eq. 7.

$$K_s q - w^2 (M_s + M_L) q = 0 \quad (7)$$

In the above expression, K_s is the stiffness matrix of shell, M_s is the mass matrix of shell, M_L is the mass matrix of fluid,

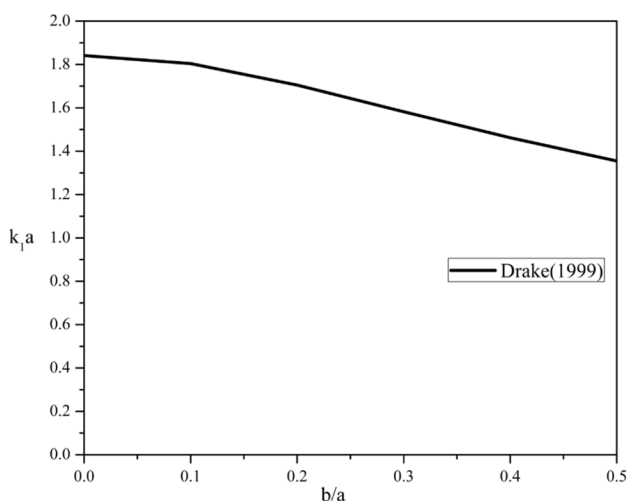


Fig. 4 Values of $k_1 a$ for a range of annular tanks

ω is the circular frequency, q is the vector of parameters of Ritz expansion.

A lot of researchers have implemented the technique of baffles in mitigating the sloshing frequency in liquid storage tanks. Biswal et al. [67] studied the influence of baffle on the dynamic response of partially filled cylindrical tank. It was concluded that the slosh frequencies of liquid in a tank were reduced by the use of the baffle and have greater influence on the slosh frequencies of liquid when placed near to the liquid free surface than toward the bottom of the tank. Gavriluk et al. [68] presented fundamental solutions of linear fluid sloshing problem in a vertical circular cylindrical tank having a thin rigid-ring horizontal baffle. An analytically oriented approach was developed to provide accurate approximations of natural frequencies and modes. The proposed method could also be used to analyze the effect of position and size of baffles within tank. The method can also be extended to solve linear sloshing problem in cylindrical tanks with two or more horizontal baffles. Panigrahy et al. [69] conducted experimental study on partially filled square tanks using shake table assembly. The effect of varying dimension and orientation of various types of baffles such as horizontal, vertical and ring type on nonlinear behavior and damping characteristics of liquid sloshing on a partially filled three-dimensional square tank was studied. It was observed that introduction of baffles decreases sloshing effect and ring baffles proved to be more effective as compared to the conventional baffles. Hasheminejad and Mohammadi [70] studied the efficiency of horizontal placed baffle in comparison with vertical bottom mounted baffle in controlling the sloshing behavior of partially filled circular tanks. Wang et al. [71–75] detailed the performance of rigid cylindrical tanks with a rigid annular baffle using analytical technique and evaluated the free vibration and seismic response of container under lateral excitation with multiple annular baffles. Xue et al. [76] studied the mitigation technique in the form of perforated baffles on reducing the sloshing effects in rectangular tanks using analytical and experimental approach. The other prominent research study in the liquid storage tanks installed with baffle systems includes the work done by Yoon and Park [77].

The study mitigated the sloshing phenomenon in circular tanks with the help of different arrangements of ring baffles. The other sloshing mitigating techniques include the installation of solid foam elements [78], curved baffles [79], slotted porous screens [80], perforated floating plates [81], moving baffles [82] along with the installation of passive and active control techniques to mitigate the sloshing effects by [83–87]. Recently, Kangda et al. [88] carried a numerical study on circular tanks with and without rigid, centrally placed circular obstruction with the help of ANSYS software to obtain sloshing parameters of circular tanks. A detailed study over wide range of circular tank geometries was

performed. Charts for obtaining sloshing mass and sloshing frequency for various geometries of circular tanks were also proposed. It is observed that with the increase in the diameter of the circular obstruction, the sloshing frequency and the percentage of liquid in sloshing mode decreases in first mode. The behavior of these sloshing parameters in higher modes was also evaluated. The studies discussing the presence of the obstruction in rectangular tanks and its effects on the sloshing parameters are limited. The graphs and equations to determine the same needs to be studied in the future. The incorporation of the above-mentioned results in the available design codes shall be helpful in determining the accurate pressure forces exerted on the tank wall and base.

Complex tank geometries

As discussed earlier, the analytical results for sloshing parameters of liquid are available for circular and rectangular tanks only. The need to construct special shape tanks results in minimizing the cost of construction as achieved by Intze tanks. As per the IITKGSDMA [89] document, during an earthquake, all the contents of water tank or reservoir on the ground will be excited into sloshing. The document provides mathematical expressions and graphs to calculate the hydrodynamic pressure forces generated on tank wall and base for rectangular and circular ground-supported tanks. The document also includes explanatory examples detailing the steps to be followed in calculating the sloshing forces and instructs the engineers to convert complex tank geometries namely Intze, spherical and truncated conical shaped tanks into an equivalent circular tank based on the studies by Joshi [90] and perform the further dynamic analysis. The performance of the elevated tanks with the conical (Intze) container shapes has been studied by Moslemi et al. [91] and results show that the available equivalent techniques proposed by design standards are in good agreement with the obtained responses. The present study illustrates some significant approaches by researchers in analyzing the slosh phenomena with complex tank geometries primarily constructed as storage reservoir over an elevated staging. It has been observed that finite element investigation has been the common tool to investigate the behavior of these tanks subjected to the slosh action generated due to external excitations. Some studies also conducted experimental test to validate the results with the available numerical studies. Damatty et al. [92] developed analytical solutions to predict the sloshing parameters for conical tanks. The results show that the experimental tests were within 15% difference when compared with the mathematical solutions. The other studies investigating sloshing behavior of conical tanks and comparing the results with the equivalent technique include

[93–95]. A few researchers [93, 96–98] employed the passive control techniques namely tuned mass dampers, base isolation and baffles to control the sloshing action in conical tanks, respectively. The elliptical tanks plagued by sloshing action were studied by [99–103] whereas [104–107] installed baffles to control the action. Spherical tanks commonly used for storing and transporting various gases, liquids and liquefied gases in chemical industry have been comprehensively investigated by [108, 109] for with and without baffle condition and validating the results with the available mechanical model. The studies also investigated the viscous damping characteristics of fluid stored in spherical tank [110], seismic analysis [111] and seismic isolation of spherical tanks [112]. Recently, researchers have also proposed that a combination of horizontal and vertical baffles proves to be a better solution to control the slosh action with numerical and experimental tests on rectangular tanks [113], the baffle placement distance proves to be a critical parameter in minimizing the sloshing phenomenon with multiple vertical baffles in rectangular tanks [114] in addition the technique of unequal baffle height is found to be more efficient than equal baffle arrangement to control violent wave height arisen due to sloshing [115] in liquified natural gas tanks. The technique of moving baffles with a spring system to mitigate sloshing of fluid in rectangular tanks using coupled Eulerian–Lagrangian technique in ABAQUS interface proved effective in dissipating the sloshing energy of fluid under harmonic and seismic excitations for a fixed baffle spring arrangement system [82]. The modern researchers have also investigated sloshing phenomena in two layered fluids namely diesel oil and water experimentally and numerically using finite difference method using multiphase model NEWTANK [116] and verified the analytical and experimental results. The validation of experimental and numerical (FEM) behavior of Newtonian and non-Newtonian fluids in stepped rectangular tanks was studied by [117]. Lately, Guruswamy and Kumar [118] developed sloshing related experimental test setup investigating the aspect ratio of rectangular tank, excitation amplitude and water depth to identify hydraulic jump and nonlinear sloshing frequency. Thus, an extensive review has been presented in the present study highlighting the sloshing phenomenon under three broad heads namely tanks without obstruction, tanks with obstruction and complex tank geometries. It is also observed that for tanks with obstruction inside the liquid the fluid–structure interaction becomes quite complex and analytical solution to this problem is difficult to obtain. For such complex fluid–structure problem, numerical approach such as finite element analysis proves to be vital. The present study is extended further to review the steps involved in the finite element technique to understand the sloshing problem with the help of the most common cylindrical and rectangular tank geometries.

Modeling using ANSYS

The finite element approach elaborating the steps to be followed for modeling and analysis of fluid–structure interaction problem is done with the help of ANSYS [119] software. The type of element selected and procedure carried for obtaining sloshing parameters using finite element approach (ANSYS software) is also discussed. The general shape function, formulation and assumptions and restriction for fluid element are also discussed. The precautions that are needed to be considered for using this element are also discussed. Moreover, while modeling the tank using only fluid element the boundary conditions applied are also discussed. General precautions and considerations while modeling is discussed eventually with criteria for selection for accurate mesh size.

Element used

In the present study, “FLUID 80” is used to model the fluid in tank. It is a 3-D contained fluid element. The fluid element is defined by eight nodes, as shown in Fig. 5, having three degrees of freedom at each node: translation in the nodal x , y and z directions, respectively. The fluid element is used to model fluids contained within vessels having no flow rate. It has a unique phenomenon of free surface effect. The free surface is effect to the element is given by special springs attached at each node of the element. These springs are in vertical direction (z -axis) and hence the fluid tank needs to be modeled with z -axis as the direction of gravity. The fluid element is particularly well suited for calculating hydrostatic pressures and fluid–solid interactions. Acceleration effects, such as in sloshing problems, as well as temperature effects, may be included. The geometry, node locations and the coordinate system for the element are shown in Fig. 5 and Table 3.

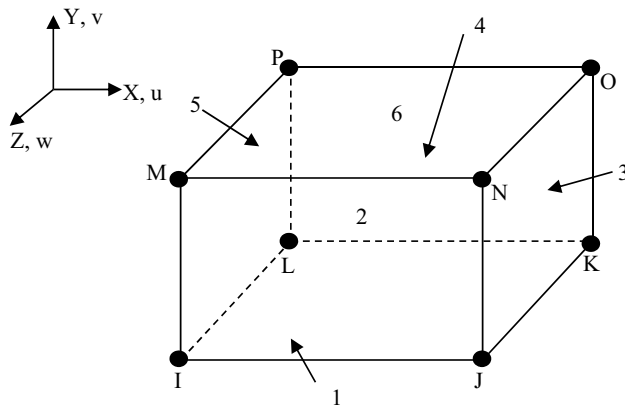


Fig. 5 Contained fluid element (FLUID80)

Table 3 Fluid 80 element input data summary

Nodes	I, J, K, L, M, N, O, P
Degree of freedom	U_x, U_y, U_z
Surface load (pressure)	Face 1 (J-I-L-K), Face 2 (I-J-N-M), Face 3 (J-K-O-N), Face 4 (K-L-P-O), Face 5 (L-I-M-P), Face 6 (M-N-O-P)
Material properties	Modulus of elasticity, density, viscosity and thermal properties such as thermal strain or Thermal coefficient of expansion

The shape functions of the 3-D FLUID80 element are expressed by Eqs. 8, 9 and 10.

$$u = \frac{1}{8} (u_I(1-s)(1-t)(1-r)) + u_J(1+s)(1-t)(1-r) + u_K(1+s)(1+t)(1-r) + u_L(1-s)(1+t)(1-r) + u_M(1-s)(1-t)(1+r) + u_N(1+s)(1-t)(1+r) + u_O(1+s)(1+t)(1+r) + u_P(1-s)(1+t)(1+r) \quad (8)$$

$$v = \frac{1}{8} (v_I(1-s)(1-t)(1-r)) + v_J(1+s)(1-t)(1-r) + v_K(1+s)(1+t)(1-r) + v_L(1-s)(1+t)(1-r) + v_M(1-s)(1-t)(1+r) + v_N(1+s)(1-t)(1+r) + v_O(1+s)(1+t)(1+r) + v_P(1-s)(1+t)(1+r) \quad (9)$$

$$w = \frac{1}{8} (w_I(1-s)(1-t)(1-r)) + w_J(1+s)(1-t)(1-r) + w_K(1+s)(1+t)(1-r) + w_L(1-s)(1+t)(1-r) + w_M(1-s)(1-t)(1+r) + w_N(1+s)(1-t)(1+r) + w_O(1+s)(1+t)(1+r) + w_P(1-s)(1+t)(1+r) \quad (10)$$

Assumptions and Restrictions of FLUID80

- Zero volume elements are not allowed.
- The element temperature is taken to be average of the nodal temperature.
- Elements may be numbered either as shown in Fig. 5 or may have the planes IJKL and MNOP interchanged.
- Structures are usually modeled with the Z-axis oriented in the vertical direction and the top surface at $Z=0.0$.
- The element may not be twisted such that element has two separate volumes. This occurs most frequent when the elements are not numbered properly.
- Elements should be rectangular (brick shaped) whenever possible, as results are known to be of lower quality for some cases using nonrectangular shapes.

- (vii) For the case of a modal analysis with irregular meshes, one can expect one or more low-frequency eigenvectors, representing internal fluid motions, without significantly affecting the vertical motion of the free surface.
 - (viii) The nonlinear transient dynamic analysis should be used instead of the linear transient dynamic analysis for this element.
 - (ix) The amount of flow permitted is limited to that which will not cause gross distortions in the element.
 - (x) The large deflection option should not be used with this element.
 - (xi) In a reduced analysis, master degree of freedom should be selected at all nodes on the free fluid surface in the direction normal to the free surface. Other master degree of freedom, if any, should only be selected normal to one or more flat planes within the fluid, with all nodes on these planes being included. Other selections may produce large internal rotations.
 - (xii) When used for a static application, the free surface must be input flat. Gravity must be input if there is a free surface. The element gives valid nodal forces representing hydrostatic pressure and also valid vertical displacements at the free surface. Other nodal displacements, which may be large, represent energy-free internal motions of the fluid.
 - (xiii) Fluid element at a boundary should not be attached directly to structural elements but should have separate coincident nodes that are coupled only in the direction normal to the interface.
 - (xiv) Arbitrarily small numbers are included to give the element some shear and rotational stability.
 - (xv) Only the lumped mass matrix is available.
- is generated by extrude command to generate the volume of rectangular tank with suitable height as required.
- (vi) While modeling the cylindrical tank, one must see that the nodes at the wall interface are rotated in the direction perpendicular to the wall or in a radial direction. This can be achieved by rotating the nodes option given in the menu (VROTAT) to develop the volume of cylindrical tank with desired height.
 - (vii) The gplot command is useful in accurate meshing of the tanks. Hexagonal mapped meshing is selected for meshing; method of edge length should be preferred instead of division of element method.
 - (viii) While modeling a liquid storage tank with curved surface, the option to be selected to avoid errors of negative pivot is choosing the springs which give the effect of free surface only on the surface. For that KEYOPT (2) needs to be given a value of 1.
 - (ix) Application of boundary condition should be followed as directed in the next section.
 - (x) For speedy analysis, no. of modes extracted should be as small as possible (probably 5 in number) and frequency range should also be closely spaced.
 - (xi) In ANSYS while performing modal analysis, the results for modal mass can be obtained by performing: GET command by using BLOCK LANCZOS method for determination of sloshing (convective) frequency, modal mass excited and hydrodynamic pressure developed at tank walls.
 - (xii) The transient (dynamic or seismic) analysis is performed by selecting “transient” tab in new analysis option followed by creating an array of time history and applying the same to the base of the structure using define load option.

Steps in modeling tanks without obstruction

- (i) Selection of structural as well as ANSYS fluid is must before modeling tanks from the preferences menu.
- (ii) Acceleration due to gravity needs to be given in vertical z-direction with free surface at $z=0$, when fluid element is used.
- (iii) The fluid properties namely bulk modulus of water is taken as $2.18 \times 10^9 \text{ N/m}^2$ and mass density as 1000 kg/m^3 .
- (iv) The “keypoints” command is used to mark the coordinates of the rectangular and circular tanks. The x and y coordinates are created for rectangular tanks whereas for circular tanks the radius and height of tank are specified using the active cs option.
- (v) The “areas” command helps in creating the planar area, i.e., rectangular or circular tank as the case may be. For rectangular tank, the volume of the tank

Steps in modeling tanks with central obstruction

- (i) The steps as mentioned in “[Steps in modeling tanks without obstruction](#)” Section must be followed to model rectangular as well as cylindrical tanks of desired dimensions without any obstructions. For tanks with obstruction, the researchers must mention the size of obstruction to develop tanks with desired obstruction in form of columns or baffles as discussed in the review.
- (ii) The designers must mention the coordinates of the obstruction along with the coordinates of tank in step iv of the previous section and they are illustrated in Fig. 6 to develop tanks with and without obstructions and follow all the steps as discussed in previous section to study the sloshing phenomenon in finite element ANSYS software. The coordinates of a sample rectangular tank are pictographically elaborated in

Fig. 6 Coordinates of rectangular tank with and without obstruction

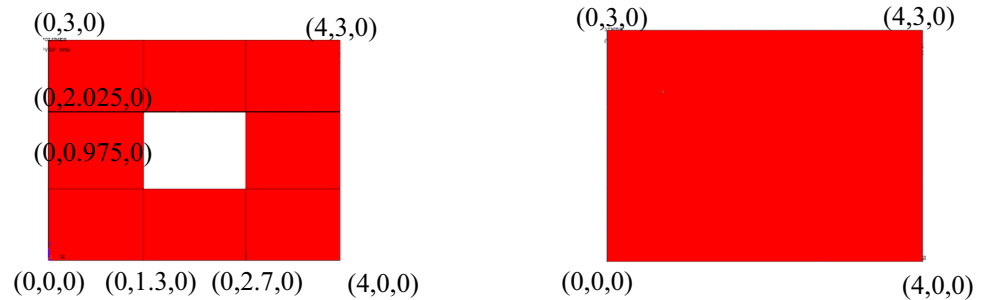
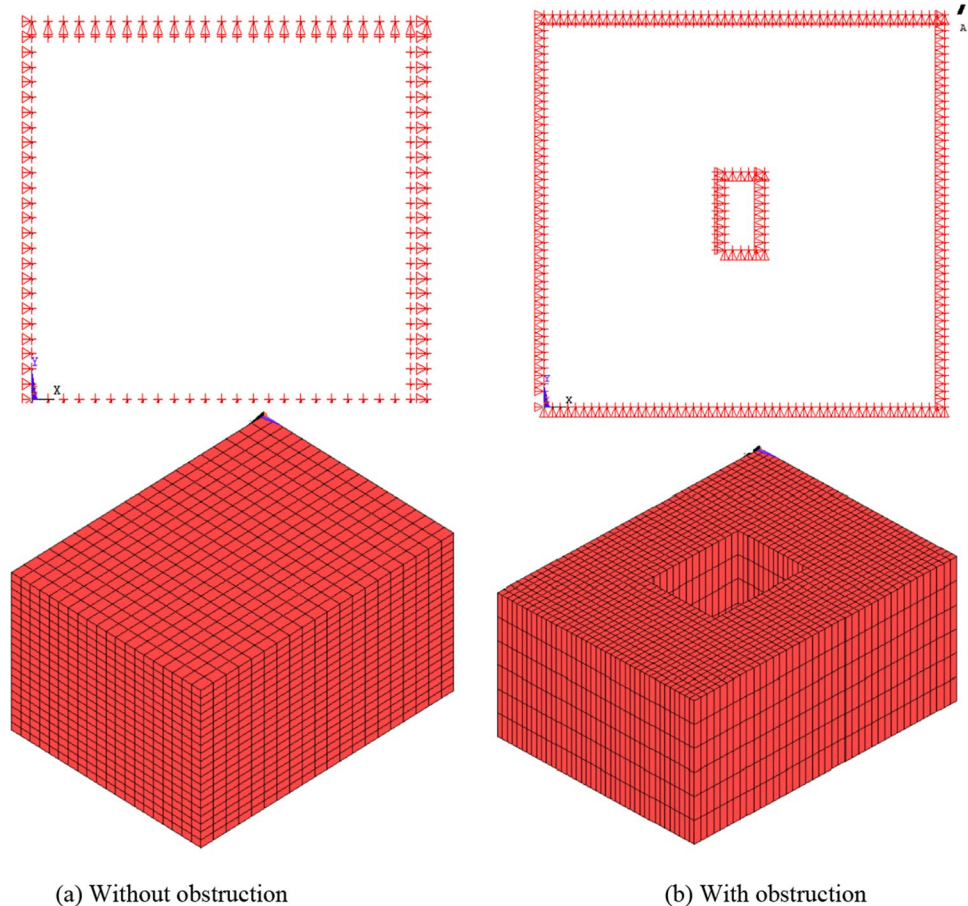


Fig. 6 along with the explanation of rectangular tank without obstruction. The coordinates must be specified in the keypoints command as per step iv of the previous section. The coordinates must follow (x,y,z) system with z coordinate taken as zero and generated by extrude command to generate the tank of desired volume of desired height. The tank in consideration is having plan dimension 4×3 m without obstruction and the same rectangular tank is having an obstruction of size 1.4×1.05 m.

Boundary condition to liquid storage tank

As mentioned earlier the study is done by using fluid element only with the basic assumption that the walls of the tank are rigid. Hence, to satisfy the condition above the boundary condition applied is such that the nodes of the fluid element at the wall interface are fixed in the direction perpendicular to the wall. For a rectangular tank, the nodes at the walls perpendicular to Cartesian x-axis are fixed in x direction, while nodes at the wall perpendicular to y-axis are fixed in y direction. The nodes at the bottom surface or the base of the tank are fixed only in vertical direction. With this, it is ensured that the nodes are not allowed past

Fig. 7 Constrained wall of rectangular tank



the tank boundaries but are free to move in other directions. Figure 7 shows the plan of a rectangular tank showing nodes and obstruction constrained in perpendicular direction. For cylindrical tank, when the tank is modeled, for all the nodes the nodal coordinate system is oriented along with the global Cartesian coordinate system. The nodes at the wall periphery do not have coordinate system such that its x -axis is not perpendicular to the curved surface. The nodal coordinate system should be rotated in a radial direction. This is done by a command in the utility menu of modeling and modify. The nodes can be rotated such that its x -axis is perpendicular to the surface area. Figure 8 shows the plan of a circular tank showing nodes and obstruction constrained in perpendicular direction.

Modal analysis using ANSYS

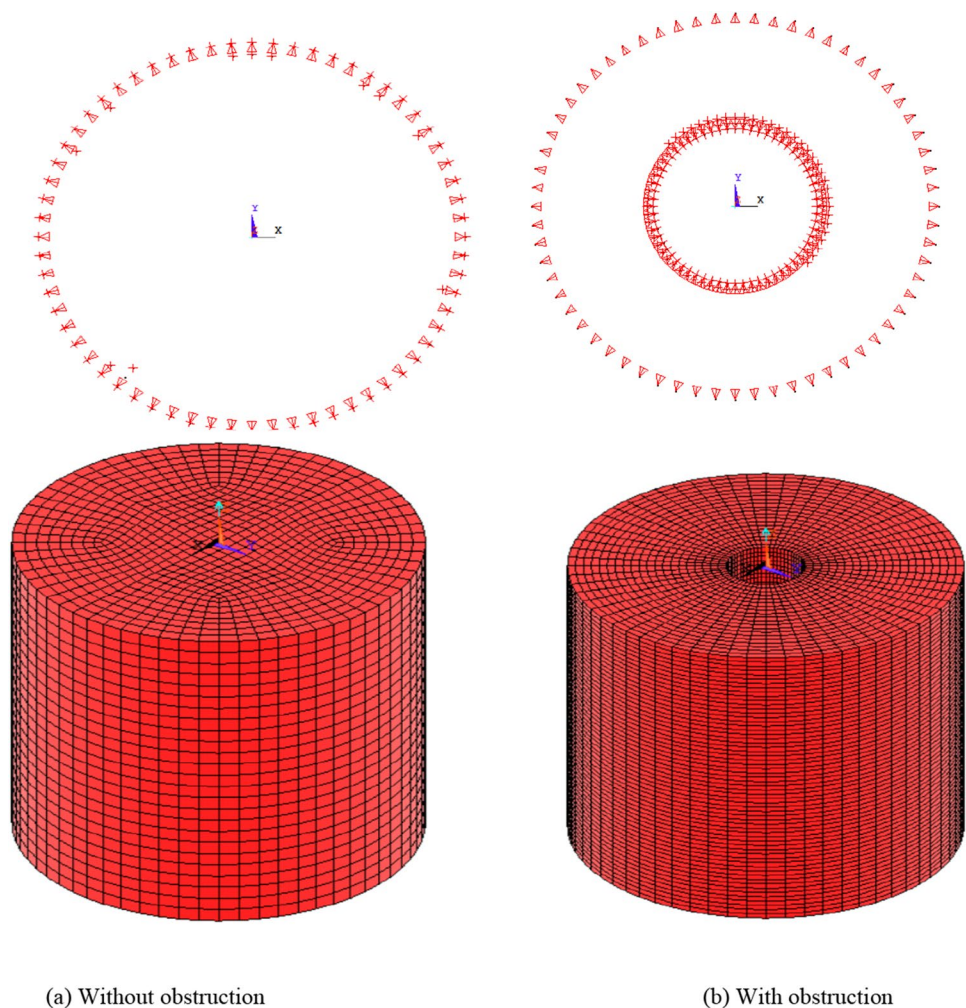
In ANSYS, various methods can be used to perform free vibration analysis of a model. But for fluid element, recommended method is reduced method. However, the method

requires selection of master nodes. The problem with the method is that if we select few master nodes and run the analysis, the results for sloshing frequency are acceptable, but the result for modal mass excited does not match with the theory. This problem can be solved when all the nodes are selected as master nodes which give good results for modal mass excited. However, if the mesh is fine and number of nodes of fluid element is large, the analysis takes a large amount of time and for very large number of nodes the analysis terminates. Hence, another method, i.e., Block Lanczos method is opted for modal analysis. Methods used for modal analysis are discussed below.

Reduced method

The reduced method is one of the available methods for modal analysis of fluid using ANSYS. The reduced method is faster because it uses reduced system matrices to calculate the solution. This method uses the HBI algorithm (Householder-Bisection-Inverse iteration) to calculate the

Fig. 8 Constrained wall of circular tank



(a) Without obstruction

(b) With obstruction

eigenvalues and eigenvectors. The advantages and disadvantage of reduced method are as follows:

Advantage

- (i) Recommended for FLUID80 with proper selection of master nodes gives results for sloshing frequency in lesser time.

Disadvantage

- (i) Does not give results for modal mass unless all the nodes of fluid element are selected as master nodes.
- (ii) If number of master nodes is too large the time of computation is large or the program terminates.

Block Lanczos method

As mentioned above reduced method which is recommended for fluid element has certain limitation in calculating modal mass. As a result, Block Lanczos method is adopted in the present study to carry out free vibration analysis. The block shifted Lanczos algorithm is a variation of the classical Lanczos algorithm, where the Lanczos recursions are performed using a block of vectors, as opposed to a single vector.

Advantages

- (i) The results for both sloshing frequency and modal mass excited both are in good agreement with the theoretical results.
- (ii) Takes lesser time even for a finer mesh.
- (iii) Gives modal mass excitation even for rotational degree of freedom.

Thus, it is highly recommended to select Block Lanczos method for evaluating sloshing parameters. The method is also helpful in predicting the hydrodynamic pressures generated by the fluid on tank wall by activating the calculating element results option. The command “TimeHist Postpro” enlists the pressure developed at different nodes and elements of the fluid element. A detailed example is followed next validating the advantages of Block Lanczos method.

Modal analysis methods in ANSYS

A rectangular tank with dimension $4 \text{ m} \times 3 \text{ m} \times 2 \text{ m}$ ($L \times B \times H$) is considered. Bulk modulus of water is taken as $2.18 \times 10^9 \text{ N/m}^2$ and mass density as 1000 kg/m^3 . Only fluid element, FLUID80, is used. Boundary condition is applied

Fig. 9 Cylindrical tank ANSYS model

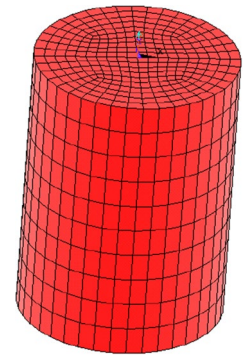


Table 4 Comparison of results for different modal analysis methods

Analysis method	1st mode			
	Sloshing frequency (Hz)		mc_1/m	
	Longer direction	Shorter direction	Longer direction	Shorter direction
NZSEE (1986)	0.42	0.50	0.49	0.38
Reduced method	0.42	0.49	–	–
Block Lanczos	0.42	0.50	0.47	0.37
PCG Lanczos	0.42	0.50	0.47	0.37

to give the effect of rigid wall as shown in Fig. 9. This is achieved by constraining the nodes at wall interface in the direction perpendicular to the wall. Different modal analysis methods are compared with each other as shown in Table 4. Thus, it is noted from the study that Block Lanczos and PCG Lanczos methods are an efficient technique determining the sloshing parameters in comparison with the reduced method as discussed in the ANSYS manual.

Effect of mesh size on sloshing parameters.

The mesh size is an important modeling parameter which affects the results for sloshing frequency and sloshing mass excited and therefore in this section circular as well as rectangular tanks without obstruction are modeled with different mesh sizes and its effect on dynamic response on these two tanks is studied.

Circular tanks

A cylindrical tank dimensions with diameter 6 m and height 5 m is considered. Bulk modulus of water is assumed as $2.18 \times 10^9 \text{ N/m}^2$ with density as 1000 kg/m^3 . The wall is assumed to be rigid and hence the nodes on the wall interface are constrained in radial direction as shown in Fig. 9. It

Table 5 First mode sloshing frequency and mass ratio of cylindrical tank for different mesh sizes

Mesh size H×D	Frequency Hz	mc ₁ /m	Time taken Sec
5×5	0.377	0.25	2
10×10	0.386	0.264	10
15×15	0.388	0.268	71
20×20	0.389	0.269	220
25×25	0.389	0.27	3420
10×5	0.386	0.261	10
5×10	0.377	0.25	3
15×5	0.388	0.264	4
5×15	0.377	0.252	17
25×10	0.389	0.269	61
10×25	0.386	0.265	110
30×30	0.389	0.271	6120
Housner (1963)	0.389	0.270	–

can be observed from Table 5 that optimum meshing of geometrical parameters (i.e., height and diameter) of cylindrical tank is important in obtaining correct results of sloshing frequency and sloshing mass of cylindrical tanks without obstruction when compared with analytical results. The time taken for analysis is noted for different mesh sizes.

The problem of meshing is addressed in this section and it is observed that fine meshing is not the ideal solution to get desired solution in the prescribed ANSYS interface and the highlighted results in Table 5 clarifies that an optimum meshing yields accurate results with available analytical solutions within desired time frame with deviation in results obtained at excessive fine meshing of tanks. Moreover, the capability of the operating system also plays a critical role in solving the finite element problems within the limited available time. Thus, a mesh size (shown in

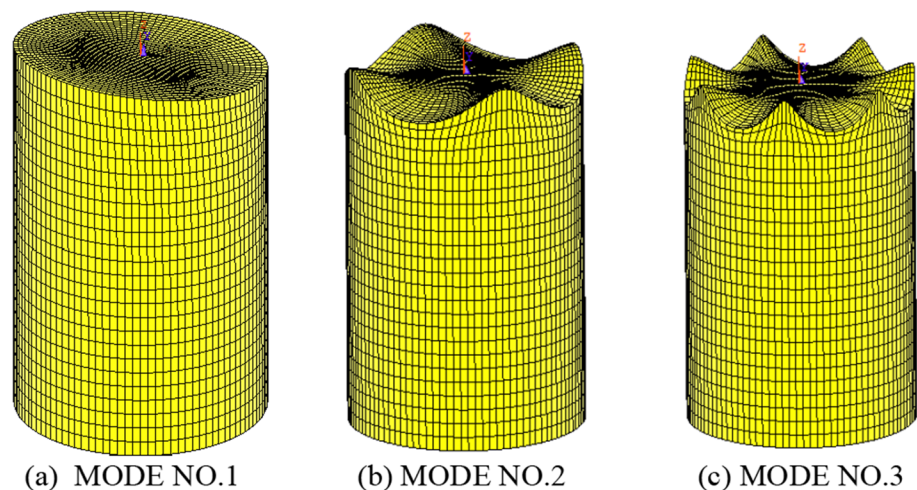
bold) of 20×20 units to 25×25 units for a tank size 5 m height and 6 m diameter is sufficient enough to obtain exact solution as obtained from analytical formulae. The mesh size of 30 x 30 units clearly depicts the over mesh size for the selected problem and results in excessive time to determine the slosh parameters. The above-mentioned observation is applied to the real cylindrical tank as studied by Haroun [16] and Stuart [18] using linear regression approach.

The modal analysis of the above-mentioned cylindrical tank is done to check the modeling and the result obtained using ANSYS. The problem considered here is from Haroun [16] which gives first three sloshing frequencies. The tank is modeled by using only fluid element, i.e., FLUID80 only with the discussed boundary conditions. The nodes at the wall surface are constrained in the direction perpendicular to the wall, i.e., in radial direction as shown in Fig. 10.

Problem Statement: A cylindrical tank dimensions with diameter 14.64 m and height 21.96 m is considered. Bulk modulus of water is taken as 2.18×10^9 N/m² with density as 1000 kg/m³. The wall is assumed to be rigid and hence the nodes on the wall interface are constrained in radial direction. Table 6 shows the comparison of sloshing frequency with analytical study obtained for a circular tank. Figure 10 shows the mode shapes of cylindrical tank. The results show the meshing size obtained from the regression

Table 6 Comparison of sloshing frequencies for cylindrical tank

Sloshing Frequency (Hz)			
Mode No	Haroun [16]	Stuart [18]	Present Study
1	0.25	0.2515	0.250
2	0.4255	0.4329	0.420
3	0.5384	0.5573	0.530

Fig. 10 Mode shapes of cylindrical tank

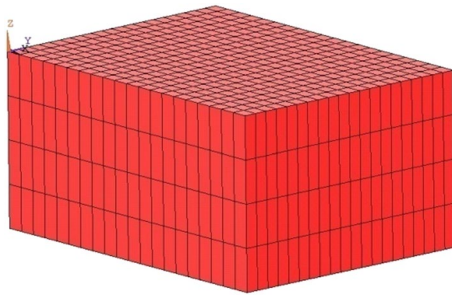


Fig. 11 Rectangular tank (4×3×2) ANSYS model

Table 7 First mode sloshing frequency and convective mass ratio of rectangular tank for different mesh sizes

Size of mesh (m) L:B:H	Longer direction		Shorter direction		Time Taken sec
	Freq Hz	mc _l /m	Freq Hz	mc _l /m	
8:6:4	0.423	0.45	0.51	0.31	1
12:9:6	0.423	0.46	0.506	0.346	3
16:12:8	0.423	0.467	0.504	0.359	6
20:20:20	0.423	0.469	0.502	0.372	38
25:25:25	0.423	0.471	0.502	0.373	90
20:20:10	0.423	0.469	0.504	0.365	5
20:10:20	0.422	0.468	0.5	0.369	17
20:18:12	0.423	0.469	0.503	0.368	14
10:20:20	0.425	0.462	0.502	0.372	9
20:18:16	0.423	0.469	0.503	0.37	23
12:10:14	0.423	0.463	0.501	0.367	1
20:22:24	0.423	0.47	0.502	0.373	43
23:21:20	0.423	0.47	0.502	0.372	68
25:16:19	0.423	0.47	0.502	0.371	35
30:30:30	0.423	0.472	0.502	0.374	300
40:40:40	0.423	0.472	0.502	0.374	3656
NZSEE (1986)	0.420	0.480	0.502	0.380	–

approach is sufficient enough to get the accurate solution and is verified with the available analytical studies.

Rectangular tank without obstruction

A tank with dimension 4 m×3 m×2 m is considered. Bulk modulus of water is taken as 2.18×10^9 N/m² and mass density as 1000 kg/m³. Boundary condition is applied to give the effect of rigid wall as shown in Fig. 11. This is achieved by constraining the nodes at wall interface in the direction perpendicular to the wall.

It can be observed from Table 7 that an optimum meshing of geometrical parameters (i.e., length, width and height) of rectangular tank is important in obtaining correct results of sloshing frequency and sloshing mass of rectangular tanks without obstruction when compared with analytical results.

The results also show that fine meshing beyond the optimum value results in greater time to evaluate the problem and leads to undesired frequency and number of iterations also increases to obtain the solution. The time taken for analysis can further be reduced if the steps directed above are followed.

A detailed investigation is further carried out to validate the results of sloshing frequency with the available analytical solutions from the obtained optimum mesh size. Four rectangular tanks with dimensions 4 m×3 m×1 m, 4 m×3 m×2 m, 4 m×3 m×3 m, 4 m×3 m×4 m are considered. Bulk modulus of water is taken as 2.18×10^9 N/m² and mass density as 1000 kg/m³. ANSYS model of rectangular tank is shown in Fig. 11. Table 8 shows the comparison of sloshing frequencies along longer and shorter directions with analytical studies. The formula given for the natural frequency for the rectangular water tank by Abramson (1966) and NZSEE (1986) is given by Eqs. 11 and 12, respectively.

$$\omega_n^2 = (2n - 1) \frac{\pi g}{a} \tanh \left[(2n - 1) \frac{\pi h}{a} \right] \quad (11)$$

$$f_i = \frac{\sqrt{\alpha_i \tanh(\alpha_i H/l)}}{2\pi \sqrt{l/g}} \quad (12)$$

where ω_n = sloshing frequency of n th mode in rad/sec. $\alpha_i = \pi/2, 3\pi/2, 5\pi/2 \dots$ a = length of tank in direction of excitation. f_i = sloshing frequency of i th mode in Hz. g = Acceleration due to gravity. h, H = height of liquid. l = half the length of wall.

The comparison of convective mass ratio for first three modes is shown in Table 9 and can be found out by analytical formula given by Abramson (1966), which is given by Eq. 13. The mode shapes of rectangular tank are as shown in Fig. 12.

$$m_n = M_l \left(\frac{8 \tanh\{2n - 1\} \pi r}{\pi^3 r (2n - 1)^3} \right) \quad (13)$$

The convective mass ratio results show that the optimum mesh size as obtained from Table 7 is suffice to validate the analytical solutions and perform further analysis and design of ground-supported liquid storage tanks using finite element method. Thus, the author concludes from the detailed investigation of mesh size that an optimum meshing of the tank dimensions is required to obtain exact solution using finite element method. The length, width, height and diameter of the rectangular and cylindrical tanks, respectively, must be divided into optimum number of parts to validate the results obtained from ANSYS package. Improper or excessive meshing of the tank dimensions results in deviated, incorrect or no solution at all with loss of time in finite element

Table 8 Sloshing frequency for different rectangular tanks

L x B x H (m)	Mode No	Frequency (Hz)					
		Longer direction			Shorter direction		
		Abramson (1966)	NZSEE (1986)	Present Study	Abramson (1966)	NZSEE (1986)	Present Study
4 × 3 × 1	1	0.36	0.36	0.36	0.45	0.45	0.45
	2	0.76	0.76	0.78	0.88	0.88	0.907
	3	0.99	0.99	1.08	1.14	1.14	1.24
4 × 3 × 2	1	0.423	0.423	0.423	0.50	0.5	0.502
	2	0.77	0.77	0.77	0.88	0.88	0.882
	3	0.99	0.99	0.99	1.14	1.14	1.14
4 × 3 × 3	1	0.44	0.44	0.44	0.51	0.51	0.51
	2	0.77	0.77	0.763	0.880	0.88	0.88
	3	0.99	0.99	0.982	1.14	1.14	1.12
4 × 3 × 4	1	0.44	0.44	0.440	0.51	0.51	0.51
	2	0.770	0.77	0.760	0.88	0.88	0.860
	3	0.99	0.99	0.960	1.14	1.14	1.07

Table 9 Convective mass ratios for different rectangular tanks

L x B x H (m)	Mode No.	m_n/M_l			
		Longer direction		Shorter direction	
		Abramson (1966)	Present Study	Abramson (1966)	Present study
4 × 3 × 1	1	0.676	0.663	0.60	0.59
	2	0.037	0.0294	0.029	0.022
	3	0.0083	0.0077	0.0063	0.0056
4 × 3 × 2	1	0.473	0.47	0.375	0.37
	2	0.019	0.018	0.014	0.0132
	3	0.004	0.004	0.0031	0.0025
4 × 3 × 3	1	0.338	0.34	0.257	0.255
	2	0.0127	0.012	0.0096	0.0088
	3	0.003	0.0024	0.002	0.0017
4 × 3 × 4	1	0.257	0.254	0.194	0.19
	2	0.0096	0.0086	0.0072	0.0063
	3	0.002	0.0016	0.0015	0.0011

package. The author advocates that the above-mentioned technique will be useful to derive the analytical sloshing parameters for rectangular tanks with obstruction in higher modes as no formulae are available as discussed for cylindrical tanks and is the future scope of work. The graphs for hydrodynamic pressure, sloshing frequency and convective mass excited can be derived for any geometry of rectangular tanks in higher modes with obstruction as obtained for cylindrical tanks with and without obstruction in the past. The author also suggests that the sloshing parameters in complex tank geometries with standard equations to obtain the same

using the above discussed finite element analysis shall help future researchers in obtaining exact sloshing parameters rather implying the present equivalent technique, i.e., converting the complex tank geometries into circular tanks of equivalent dimensions. The present review and modeling aid shall help researchers in modeling and analyzing liquid storage tanks of any shape and size with variable and multiple obstructions as in the present study steps to model central obstruction is discussed and can be extended to model tank of any shape with multiple obstructions.

In the above equation, n = slosh mode, m_n = mass of liquid acting in that mode, $r = h/a$, h = height of water in the tank, a = length of tank in direction of excitation. M_l = Total mass of the water in tank.

Conclusion and discussion

The concept of sloshing was first introduced in the year 1933 by Westergaard [120] but significant contribution has been done only in the very recent past. The analytical results for sloshing parameters of liquid are available for circular and rectangular tanks only. For tanks of other shapes and for tanks with obstruction inside the liquid the fluid–structure interaction becomes quite complex and analytical solution to this problem is difficult to obtain. For such complex fluid–structure problem, numerical approach such as finite element analysis is used. The study is concluded with the following important conclusions as illustrated as follows:

- (1) An attempt has been made to outline the literature review in the area of sloshing due to forced sinusoidal and seismic excitations. From the literature review, it

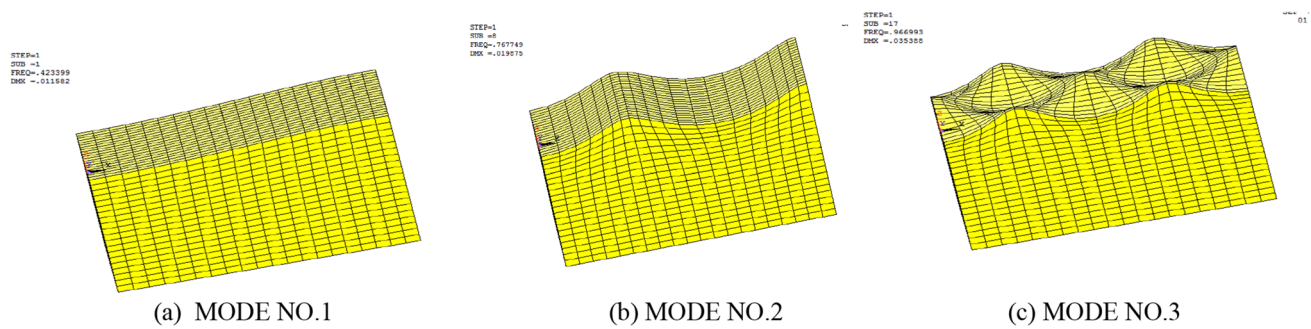


Fig. 12 Mode shapes of rectangular tank

has been observed that most of the research works in the area of slosh dynamics was carried out in the area of numerical and analytical simulations with limited experimental tests.

- (2) The overall observation from these experimental results gives the idea that sloshing in a tank is a function of the liquid depth, the dimensions of the tank, the amplitude and frequency of excitation and density of the liquid. Also, the suppression of sloshing behavior using baffles has also been a subject of interest in the recent years. The subject of sloshing is associated with various complexities because of the nonlinearities of its behavior yet it is one of the most untapped areas of research.
- (3) The present study also discusses the basic theory and methods of modeling and analysis in ANSYS are covered. By using fluid element, an apt modal analysis technique, i.e., Block Lanczos method is selected from the available methods and the results are found out to be in good agreement with the classical approach for the results like frequency, mode shape.
- (4) Modeling of cylindrical as well as rectangular liquid storage tanks is done and analyzed. The results obtained are satisfactory and consistent with the theory. Effect of meshing on the sloshing parameters is also studied in depth.
- (5) It is proved that obtaining an optimum mesh size results in more accurate validation of results with classical method within limited time frame whereas an excessive (fine) mesh size deviates the sloshing parameters with an enormous time required for analysis. Moreover, the problem of fine mesh also results in limited number of modes extracted from analysis and further gives inaccurate solutions. The results are compared with the available analytical approaches.
- (6) The study also directs the future researchers modeling and analysis aid to study the problem of sloshing in complex tanks geometries with and without obstructions as future scope of work. The steps specified in the present study with help in modeling baffles and study

the effect of different shapes of obstruction placed at an eccentric location in rectangular and circular tanks on the sloshing phenomenon.

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Declarations

Conflict of interests The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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